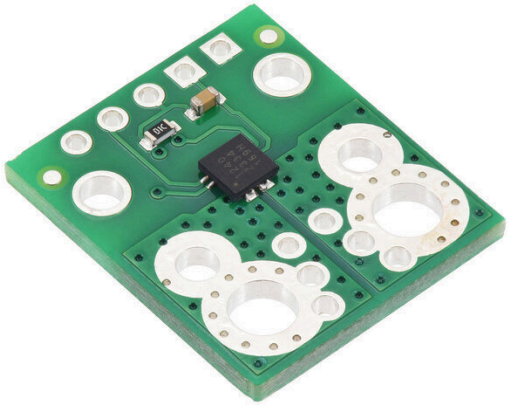
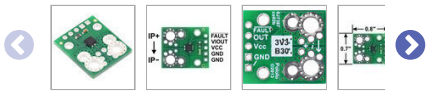
ACS71240KEXBLT-030B3 Current Sensor Carrier -30A to +30A, 3.3V





Pololu item #: 5241
Brand: Pololu

216 in stock
supply outlook
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Status: Active and Preferred
RoHS 3
Free add-on shipping in USA
Free shipping in USA over $75

Price break	Unit price (US$)
1	5.25
5	4.83
25	4.44
100	4.09

Quantity: 1
backorders allowed

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This board is a simple carrier of Allegro's **ACS71240KEXBLT-030B3** Hall effect-based, electrically isolated current sensor, which offers a low-resistance (~0.6 mΩ) IC current path and a 120 kHz bandwidth with <5 µs response time.

Part Suffix	Range	Supply Voltage	Sensitivity	Zero Point	Fault Trip Level
030B3	±30 A (bidirectional)	3.0 V to 3.6 V	44 mV/A	1.65 V	±30 A

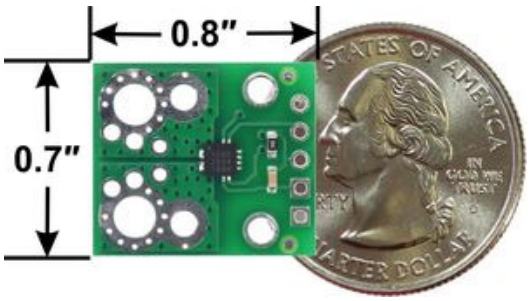
Alternatives available with variations in these parameter(s): current range **Select variant…**

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Description **Specs (12)** **Pictures (11)** **Resources (6)** FAQs (0) On the blog (0) **Distributors (60)**

Overview

We are offering these breakout boards with support from Allegro Microsystems as an easy way to use or evaluate their ACS71240 Hall effect-based, electrically isolated current sensors with overcurrent fault output; we therefore recommend careful reading of the **ACS71240 datasheet** before using this product. The following list details some of the sensor's key features:

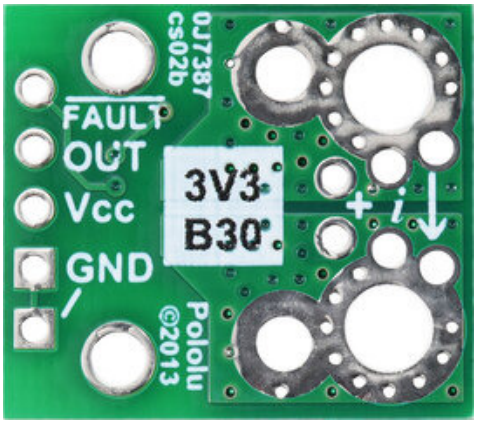


- Hall effect-based sensor with electrically isolated current path allows the sensor to be inserted anywhere along the current path and to be used in applications that require electrical isolation.
- Differential Hall sensing rejects common-mode fields, so the orientation of the sensor relative to uniform external magnetic fields (e.g. the Earth's magnetic field) has less effect on the measurement.
- The conductive path internal resistance is typically 0.6 mΩ, and the PCB is made with 2-oz copper, so very little power is lost in the module.

- 120 kHz bandwidth analog output voltage proportional to AC o
- Less than 5 µs response time.
- Output is not ratiometric (i.e. the zero point and sensitivity are which provides immunity from noisy supplies.
- Overcurrent fault output with 1.5 µs response time indicates sensing range and can be used for fast short-circuit detection.
- Integrated digital temperature compensation circuitry allov temperature in an open loop sensor.
- Automotive-grade operating temperature range of -40°C to 12
- 0.7″×0.8″ carrier board offers a variety of ways to insert it i (breadboard-compatible) power, ground, and output pins.
- 3.3V and 5V versions available.
- Unidirectional and bidirectional versions available.

The pads are labeled on the bottom silkscreen. The silkscreen also shows the direction that is interpreted as positive current flow via the **+i** arrow.

Details for item #5241



ACS71240KEXBLT-030B3 Current Sensor Carrier -30A to +30A, 3.3V, bottom view.

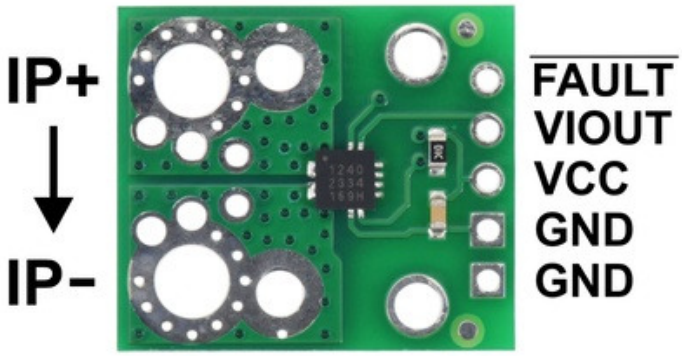


ACS71240 Current Sensor Carrier.

This carrier features the ACS71240KEXBLT-030B3, which is intended for operation around 3.3 V and is designed for bidirectional input current from -30 A to +30 A. This version can be visually distinguished from the other versions by the “3V3 B30” printed on the bottom side, as shown in the left picture above.

Part Suffix	Range	Supply Voltage	Sensitivity	Zero Point	Fault Trip Level
030B3	±30 A (bidirectional)	3.0 V to 3.6 V	44 mV/A	1.65 V	±30 A

Using the sensor



This sensor has five required connections: the input current (IP+ and IP-), logic power (VCC and GND), and the sensor output (VIOUT).

The sensor requires a supply voltage of 3.0 V to 3.6 V to be connected across the VCC and GND pads, which are labeled on the bottom silkscreen. The sensor outputs an analog voltage on VIOUT that is centered at 1.65 V and changes by 44 mV per amp of input current, with positive current increasing the output voltage and negative current decreasing the output voltage:

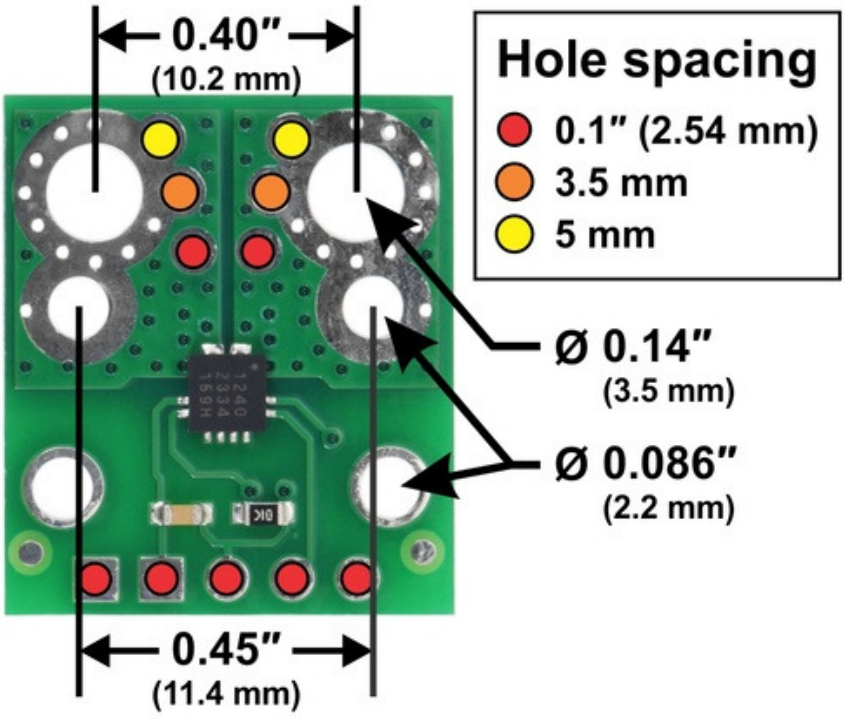
$$V_{\mathrm{IOUT}} = 1.65\mathrm{V} + 0.044\frac{\mathrm{V}}{\mathrm{A}} \cdot I_{\mathrm{P}}$$

$$I_{\mathrm{P}} = \frac{V_{\mathrm{IOUT}} - 1.65\mathrm{V}}{0.044\frac{\mathrm{V}}{\mathrm{A}}} = (V_{\mathrm{IOUT}} - 1.65\mathrm{V}) \cdot 22.7\frac{\mathrm{A}}{\mathrm{V}}$$

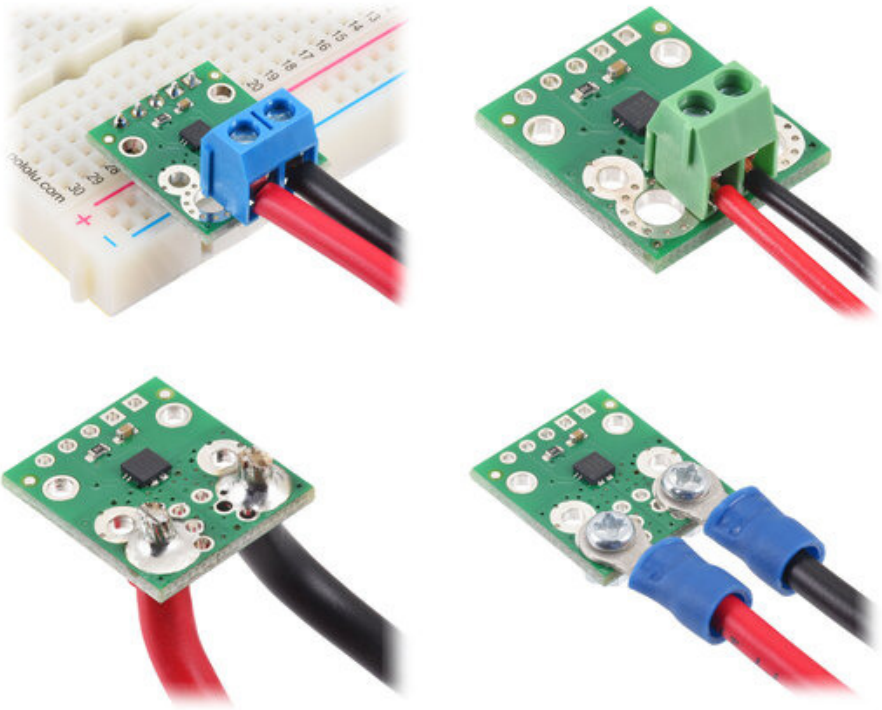
The output is not ratiometric, so the zero point and sensitivity are independent of the actual supply voltage.

The optional $\overline{\mathrm{FAULT}}$ pin is normally at VCC and is pulled low when the IP current magnitude exceeds 30 A in either direction. This pin only asserts while the fault condition is present (it is **not** latched).

The $\overline{\mathrm{FAULT}}$, VIOUT, VCC, and GND pins work with **0.1″-pitch header pins** and are compatible with standard **solderless breadboards**.

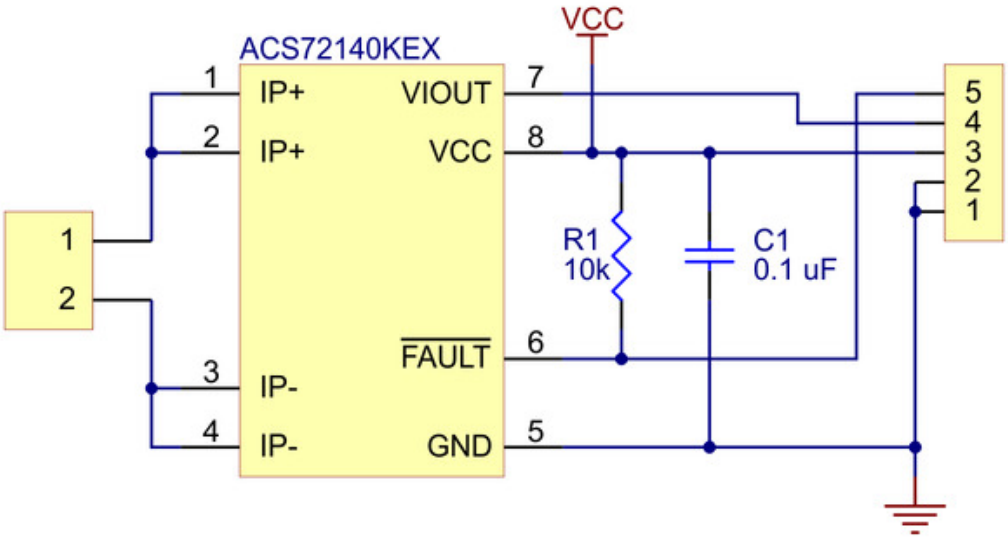


You can insert the board into your current path in a variety of ways. Holes with 0.1″, 3.5 mm, and 5 mm spacing are available as shown in the diagram above for connecting **male header pins** or **terminal blocks**. For high-current applications, you can solder wires directly to the through-holes that best match your wires, or you can use solderless ring terminal connectors. The largest through-holes are big enough for 8 AWG wires or #6 or M3.5 screws, and the second-largest through-holes (and mounting holes) are sized for 12 AWG wires or #2 or M2 screws. The pictures below show some of the possible connection options:



Warning: This product is intended for use below 30 V. Working with higher voltages can be extremely dangerous and should only be attempted by qualified individuals with appropriate equipment and experience.

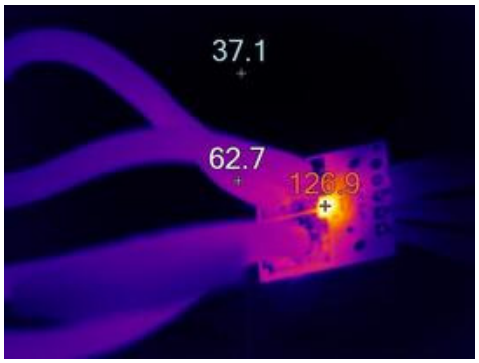
Schematic and dimension diagrams



ACS72140 current sensor carrier schematic diagram.

The dimension diagram is available as a **downloadable PDF** (274k pdf).

Real-world power dissipation considerations



Thermal image of a high-current test of a Pololu current sensor carrier (not necessarily this product).

Depending on the version, the ACS71240 can measure up to ±50 A. However, the sensor chip will typically overheat at lower currents. In our tests, we found that our ACS71240 carrier board could conduct about 45 A continuously without reaching the thermal limit for the IC. Our tests were conducted at approximately 25°C ambient temperature with no forced air flow.

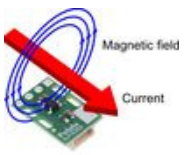
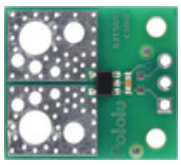
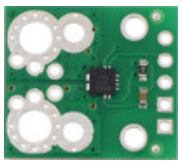
The actual current you can pass through the sensor will depend on how well you can keep it cool. The carrier's printed circuit board is designed to help with this by drawing heat out of the sensor chip. Solid connections to the current path pins (such as with thick soldered wires or large, tightly-secured lugs) can also help reduce heat build-up in the sensor and carrier board.

Warning: Exceeding temperature or current limits can cause **permanent damage** to the sensor. If you are measuring an average continuous current greater than 30 A, we strongly recommend that you monitor the sensor's temperature and look into additional cooling if necessary.

This product can get **hot** enough to burn you long before the chip overheats. Take care when handling this product and other components connected to it.

Comparison of the Pololu current sensor carriers

We have a variety of current sensors available with different ranges, sensitivities, and features. The table below summarizes our selection of active and preferred options:

	 **CT220 Contactless Current Sensor Carriers**	 **ACS3704x Current Sensor Micro Carriers**	 **ACS3704x Current Sensor Compact Carriers**	 **ACS711 Current Sensor Carriers**	 **ACS7 Sen**
Allegro Sensor	CT220xMV-HS5	ACS3704x		ACS711KEXT	
Sensing technology	XtremeSense™ TMR (tunneling magnetoresistance)	Hall effect		Hall effect	
Logic voltage range	2.7–5.5 V	3.3V versions: 3.0–3.6 V 5V versions: 4.75–5.5 V		3.0–5.5 V	3.3V 5V v
Family current range	10–800 A	10–30 A		15.5–31 A	
Range/sensitivity of individual versions	(2) **±1.5 mT / 1500 mV/mT** **±15 mT / 150 mV/mT**	ACS37041: 3.3V Bidirectional: **±10 A / 132 mV/A** **±30 A / 44 mV/A** 5V Bidirectional: **±10 A / 200 mV/A** **±30 A / 66.7 mV/A**	ACS37041: 3.3V Bidirectional: **±10 A / 132 mV/A** **±30 A / 44 mV/A** 5V Bidirectional: **±10 A / 200 mV/A** **±30 A / 66.7 mV/A**	Bidirectional:(1) **±15.5 A / 90 mV/A** **±31 A / 45 mV/A**	3.3V **±10** **±30** **±50** 5V **±10** **±30** **±50**

		ACS37042: 3.3V Bidirectional: **±10 A / 132 mV/A** **±30 A / 44 mV/A** 5V Bidirectional: **±10 A / 200 mV/A** **±30 A / 66.7 mV/A**	ACS37042: 3.3V Bidirectional: **±10 A / 132 mV/A** **±30 A / 44 mV/A** 5V Bidirectional: **±10 A / 200 mV/A** **±30 A / 66.7 mV/A**		5V U **0–50**
Max bandwidth	30 kHz	125 kHz		100 kHz	
IC current path resistance	N/A (contactless)	1.6 mΩ		0.6 mΩ	
IC isolation rating[6]	N/A (contactless)	ACS37041: 100 V_{RMS} ACS37042: 285 V_{RMS}		100 V_{pk}	
Minimum PCB creepage[7]	N/A (contactless)	ACS37041: 1.6 mm ACS37042: 2.0 mm	ACS37041: 1.6 mm ACS37042: 3.0 mm	0.75 mm	
PCB	N/A (contactless)	2 layers, 1-oz copper	2 layers, 2-oz copper	2 layers, 2-oz copper	2
Size	0.4″ × 0.62″	0.3″ × 0.4″	0.7″ × 0.8″	0.7″ × 0.8″	0
Overcurrent fault output	✔			✔	
Common-mode field rejection		✔			
Nonratiometric output		✔			
1-piece price	$4.95	$4.15 – $4.69	$4.45 – $4.99	$4.85	

Note 1: Sensitivity when Vcc = 3.3 V; actual sensitivity is ratiometric (i.e. it is proportional to Vcc).

Note 2: Sensitivity when Vcc = 5 V; actual sensitivity is ratiometric (i.e. it is proportional to Vcc).

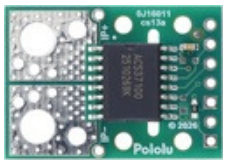
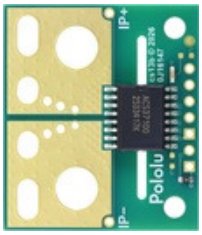
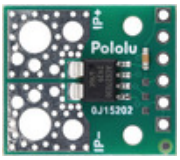
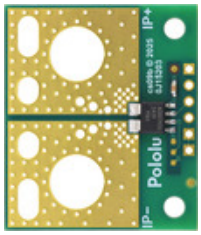
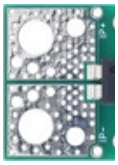
Note 3: Bandwidth can be reduced by adding a filter capacitor.

Note 4: 50A version of this carrier uses 4-oz copper PCB; all other versions of this carrier use 2-oz copper.

Note 5: 50A and higher versions of this carrier use a 4-layer PCB; all other versions of this carrier use a 2-layer PCB.

Note 6: IC component rating per manufacturer datasheet.

Note 7: Minimum creepage along PCB surface based on layout design only. Other creepage distances, e.g. along the body of the component, may be lower.

	 **ACS37100 Current Sensor Compact Carriers**	 **ACS37100 Current Sensor Large Carriers**	 **ACS37030LZ Current Sensor Compact Carriers**	 **ACS37030LZ Current Sensor Large Carriers**	 **ACS37 Current Compact**
Allegro Sensor	ACS37100		ACS37030LLZ		
Sensing technology	XtremeSense™ TMR (tunneling magnetoresistance)		Hall effect + inductive cc		
Logic voltage range	3.0–3.6 V		3.0–3.6 V		
Family current range	25–50 A		20–65 A		

Range/sensitivity of individual versions	3.3V Bidirectional: **±25 A / 52.8 mV/A** **±50 A / 26.4 mV/A**	3.3V Bidirectional: **±25 A / 52.8 mV/A** **±50 A / 26.4 mV/A**	3.3V Bidirectional: **±20 A / 66 mV/A** **±40 A / 33 mV/A** **±65 A / 20.3 mV/A**	3.3V Bidirectional: **±40 A / 33 mV/A** **±65 A / 20.3 mV/A**	3.3V Bidir **±25 A / 5** **±40 A / 3** **±65 A / 2**
Max bandwidth	10 MHz		5 MHz		
IC current path resistance	1.2 mΩ		0.7 mΩ		
IC isolation rating(4)	1097 V_{RMS}		840 V_{RMS}		
Minimum PCB creepage(5)	9.3 mm	10.1 mm	4.1 mm		11.7
PCB	2 layers, 2-oz copper	6 layers, 2-oz copper	2 layers, 2-oz copper	6 layers, 2-oz copper	2 lay 2-oz c
Size	0.8″ × 1.1″	1.4″ × 1.2″	0.7″ × 0.8″	1.4″ × 1.2″	0.8″ ×
Overcurrent fault output	✔ (configurable threshold)				
Common-mode field rejection	✔		✔		
Nonratiometric output	✔		✔		
1-piece price	$14.95	$19.95	$8.95	$11.95	$12

Note 1: Sensitivity when Vcc = 3.3 V; actual sensitivity is ratiometric (i.e. it is proportional to Vcc).

Note 2: Sensitivity when Vcc = 5 V; actual sensitivity is ratiometric (i.e. it is proportional to Vcc).

Note 3: 50A and higher versions of this carrier use a 4-layer PCB; all other versions of this carrier use a 2-layer PCB.

Note 4: IC component rating per manufacturer datasheet.

Note 5: Minimum creepage along PCB surface based on layout design only. Other creepage distances, e.g. along the body of the component, may be lower.

You can also use the following selection box to see these options sorted by current range:

Alternatives available with variations in these parameter(s): current range **Select variant…**